



# System Administration Essentials

Take charge

**S**ystem administration is the backbone of modern computing, an invisible yet indispensable force that ensures the seamless operation of software and hardware in unison. Whether managing a single workstation or an enterprise-grade server cluster, a system administrator's role is to maintain security, efficiency, and stability. This chapter explores essential administrative tasks, from user management to package handling and system monitoring.

## User and Group Management

Users and groups are fundamental to system access and permissions. Proper user management ensures security, resource allocation, and operational efficiency.

### Creating and Modifying Accounts (`useradd`, `passwd`)

Creating user accounts is one of the first steps in configuring a new system. The `useradd` command allows administrators to create new users, while `passwd` assigns them a password:

```
sudo useradd -m -s /bin/bash username
sudo passwd username
```

Here, `-m` creates a home directory, and `-s` assigns a shell.

To modify an existing user, use `usermod`:

```
sudo usermod -aG sudo username # Adds user
to sudo group
```

Deleting users and their data can be done with:

```
sudo userdel -r username
```